

- 4 The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} a & b^2 \\ c^2 & a \end{pmatrix}$, where a, b, c are real constants and $b \neq 0$.

(a) Show that $\mathbf{M}$ does not represent a rotation about the origin. [2]

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(b) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by $\mathbf{M}$. [5]

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It is given that **M** represents the sequence of two transformations in the x – y plane given by an enlargement, centre the origin, scale factor 5 followed by a shear, x -axis fixed, with $(0, 1)$ mapped to $(5, 1)$.

- (c) Find **M**. [3]

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- (d) The triangle DEF in the x – y plane is transformed by **M** onto triangle PQR .

Given that the area of triangle DEF is 12 cm^2 , find the area of triangle PQR . [2]

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